

Assignment 1 - Probability Solutions

All calculations use the values in Assignment1-1.pdf. Probabilities are exact until the final rounding step. Entropy is measured in bits using log base 2.

Question 1 - Independent events

Given $P(A) = 0.4$ and $P(B) = 0.3$, with A and B independent:

(a) Intersection

$$P(A \cap B) = P(A)P(B) = 0.4 \times 0.3 = \mathbf{0.12}.$$

(b) Union

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.3 - 0.12 = \mathbf{0.58}.$$

The intersection is subtracted because it is counted in both $P(A)$ and $P(B)$.

Question 2 - Testing independence

Given $P(A) = 0.5$, $P(B) = 0.4$, and $P(A | B) = 0.7$.

For independence, $P(A | B)$ must equal $P(A)$ when $P(B) > 0$. Here $0.7 \neq 0.5$, so **A and B are not independent**.

Equivalently, $P(A \cap B) = P(A | B)P(B) = 0.7 \times 0.4 = 0.28$, whereas $P(A)P(B) = 0.5 \times 0.4 = 0.20$. These differ.

Question 3 - Bayes' rule

Given $P(A) = 0.6$, $P(B | A) = 0.5$, and $P(B | A^c) = 0.2$.

$$P(A^c) = 1 - 0.6 = 0.4.$$

By total probability:

$$P(B) = P(B | A)P(A) + P(B | A^c)P(A^c) = 0.5 \times 0.6 + 0.2 \times 0.4 = 0.30 + 0.08 = 0.38.$$

By Bayes' rule:

$$P(A | B) = P(B | A)P(A) / P(B) = 0.30 / 0.38 = 15/19 \approx \mathbf{0.78947 (78.95\%)}$$

Question 4 - Disease given a positive test

Let D mean having the disease and + mean a positive test.

$P(D) = 0.02$; $P(D^c) = 0.98$; $P(+ | D) = 0.95$. The true negative rate is 0.90, so $P(+ | D^c) = 1 - 0.90 = 0.10$.

$$P(+) = P(+ | D)P(D) + P(+ | D^c)P(D^c) = 0.95 \times 0.02 + 0.10 \times 0.98 = 0.019 + 0.098 = 0.117.$$

$$P(D | +) = P(+ | D)P(D) / P(+) = 0.019 / 0.117 = 19/117 \approx \mathbf{0.16239 (16.24\%)}$$

For example, among 10,000 people at these rates, 200 have the disease and 190 of them test positive. Of the 9,800 healthy people, 980 test positive. Therefore, $190 / (190 + 980) = 16.24\%$ of positive tests come from people with the disease. The low prior prevalence explains this result.

Question 5 - Expected value, variance, and sample mean

x	P(X = x)	x P(X = x)	x ² P(X = x)
85	0.375	31.875	2709.375
90	0.375	33.750	3037.500
95	0.125	11.875	1128.125
100	0.125	12.500	1250.000
Total	1.000	90.000	8125.000

(a) $E[X] = \sum x P(X = x) = 31.875 + 33.750 + 11.875 + 12.500 = \mathbf{90 \text{ points}}$.

(b) $E[X^2] = 8125$.

$\text{Var}(X) = E[X^2] - (E[X])^2 = 8125 - 90^2 = 8125 - 8100 = \mathbf{25 \text{ points}^2}$.

(c) The sample is {85, 90, 85, 95, 90, 85, 100, 90}.

Sample mean = $(85 + 90 + 85 + 95 + 90 + 85 + 100 + 90) / 8 = 720 / 8 = \mathbf{90 \text{ points}}$.

The sample mean equals $E[X]$ here because the observed frequencies are 3/8, 3/8, 1/8, and 1/8, exactly matching the given probabilities. A different random sample need not have a mean exactly equal to $E[X]$.

Question 6 - Entropy

(a) For probabilities 0.4, 0.3, 0.2, and 0.1:

$H(X) = -\sum p(x) \log_2 p(x) = -[0.4 \log_2(0.4) + 0.3 \log_2(0.3) + 0.2 \log_2(0.2) + 0.1 \log_2(0.1)] = 0.528771 + 0.521090 + 0.464386 + 0.332193 \approx \mathbf{1.84644 \text{ bits}}$.

(b) With four equally likely messages, each probability is 1/4:

$H(X) = -4 \times (1/4) \times \log_2(1/4) = -1 \times (-2) = \mathbf{2 \text{ bits}}$.

The uniform distribution has the maximum entropy for four possible messages.